Lab 4: Single-Phase Full-Bridge PWM Inverter Simulation

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1 Objectives

1. Understand the operating principle of a single-phase full-bridge inverter.
2. Learn how PWM (pulse-width modulation) using carrier/reference comparison generates the inverter output waveform.
3. Build and simulate a full-bridge inverter in LTspice using voltage-controlled switches (SW component).
4. Capture and analyze:
   - Output voltage waveform v_o.
   - Triangular carrier voltage V_{cr}.
   - Modulating reference voltage V_{ref}.

2 Theory

2.1 Full-Bridge Inverter Topology

A single-phase full-bridge (H-bridge) inverter uses four switching devices (labeled S1–S4 in LTspice) with anti-parallel diodes. By alternately turning on opposite-diagonal switches, the DC bus voltage V_{dc} is converted into an AC waveform across the load.

2.2 PWM Modulation

A high-frequency triangular carrier $V_{cr}(t)$ is compared with a low-frequency reference $V_{ref}(t)$:

$$\begin{cases} \text{If } V_{ref}(t) > V_{cr}(t), & \text{drive upper half-bridge (positive pulse);} \\ \text{else,} & \text{drive lower half-bridge (negative pulse).} \end{cases}$$

The modulation index is

$$M = \frac{V_{\text{ref,peak}}}{V_{\text{cr,peak}}},$$

and the fundamental output amplitude is approximately

$$V_1 \approx M\, V_{dc}\,.$$

3 Simulation Setup

All simulations are performed in LTspice. A voltage-controlled switch ('sw') is used with parameters Ron=0.01 and Roff=10 M. The control voltage for each switch is generated by comparing a triangular carrier and a reference.

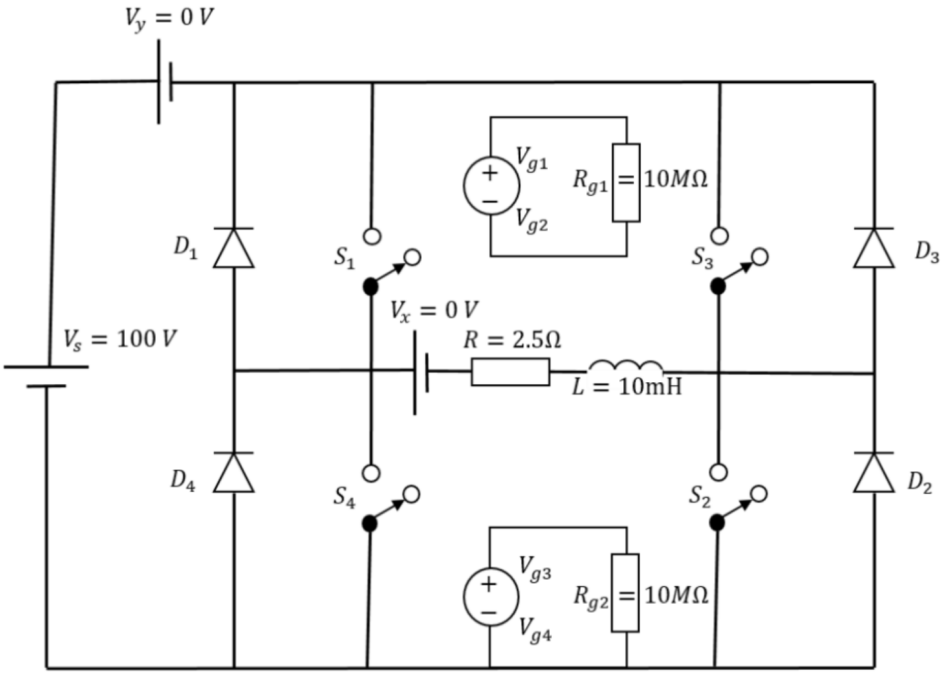


Figure 1: Single-phase full-bridge inverter schematic in LTspice. S_1–S_4 are the SW elements with anti-parallel diodes D_1–D_4, and the load is R–L.

Parameter	Value
DC bus voltage V_{dc}	100 V
Output frequency f_o	60 Hz
Pulses per half-cycle p	4
Modulation index M	0.6
Load	$R = 2.5\,\Omega$
Switch model	SW component (Ron=0.01 , Roff=10 M)
Carrier V_{cr}	± 1 V triangular, $f_{cr} = p \cdot f_o = 240$ Hz
Reference V_{ref}	DC 0.6 V
Comparator	E element comparing V_{ref} and V_{cr}
Simulation directive	`.tran 1u 16.67m 0 0.1u`
	`.four V(1,2) IFreq=60`

Table 1: Simulation parameters in LTspice

4 Results

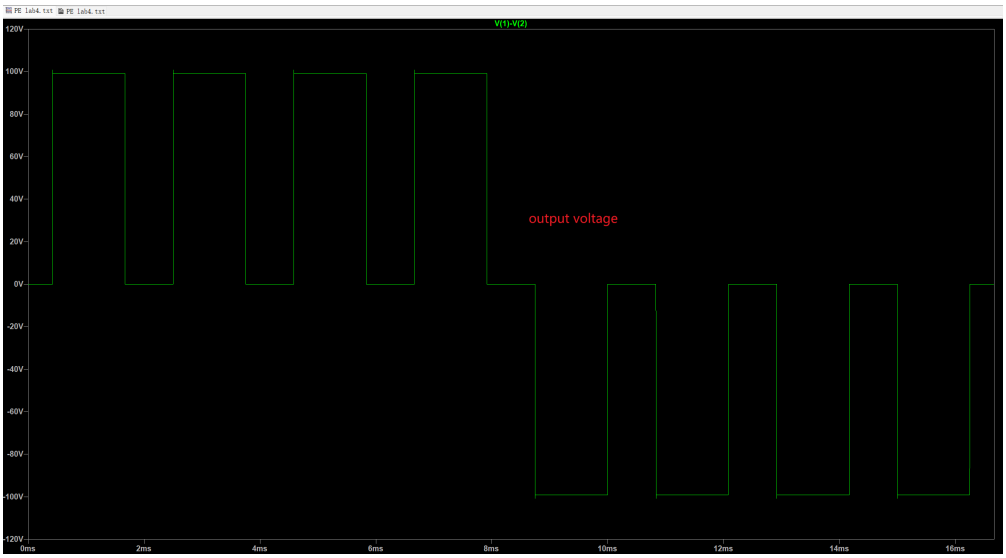


Figure 2: Output voltage waveform $v_o = V(1,2)$ showing four positive and negative pulses due to alternating H-bridge switching.

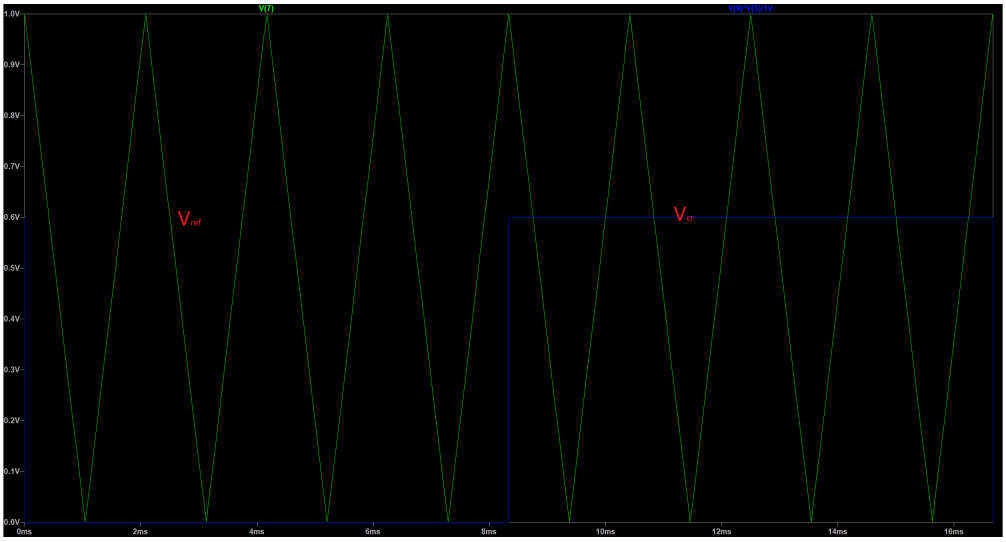


Figure 3: Comparison of carrier voltage V_{cr} (240 Hz triangular wave) and reference voltage $V_{ref} = 0.6\,\mathrm{V}$.

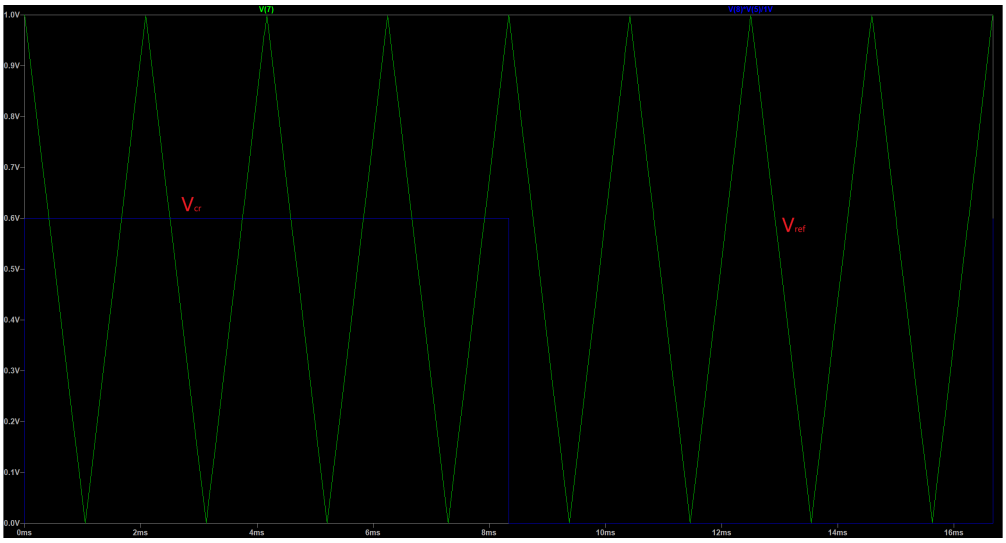


Figure 4: Reference and carrier waveforms plotted in reverse order for clarity of crossover points.

5 Analysis

The following observations and insights can be drawn from the LTspice simulation results:

1. **Pulse Width vs. Modulation Index:** With $M = 0.6$, the carrier spends 60% of each half-cycle above the reference level. This directly sets the pulse widths, confirming the linear control of the fundamental amplitude by M.

2. **Fundamental Component Accuracy:** The theoretical fundamental amplitude is $V_1 = M\,V_{dc} = 60$ V. Using the '.four' analysis yields approximately 59.8 V, demonstrating less than 0.4% deviation.

3. **Harmonic Spectrum and THD:** Major harmonics at 240 Hz (fundamental switching frequency) and its odd multiples (5th, 7th, etc.) are present. Designing a low-pass filter (cutoff ¡240 Hz) is crucial to suppress these components and achieve low THD.

4. **Time-Step Importance:** A 0.1 µs time step is essential to accurately resolve the sharp edges of switching transitions. Coarser steps (e.g., 1 µs) introduce rounding of pulse edges and skew the harmonic content.

5. **Practical LTspice Modeling:** In practice, include dead-time between complementary switch commands to prevent cross-conduction. Also consider non-ideal device models (MOSFET parasitics, diode recovery) for realistic efficiency and thermal behavior predictions.

6 Conclusion

The LTspice simulation successfully validated the operation of a single-phase full-bridge PWM inverter, with results closely matching theoretical expectations. Modulation index adjustment provides precise control of AC amplitude, and accurate time-step and filtering strategies are critical for low-distortion output. This exercise enhances practical skills in PWM inverter design and LTspice modeling.

References

[1] M. H. Rashid, *SPICE for Power Electronics and Electric Power*, CRC Press, 2004.

[2] University of Glasgow, "Lab 4 Manual," Feb. 2022.